

AHMED MAGHA

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SUMMARY

AI & Machine Learning Engineer with expertise in Data Science, Intelligent Document Processing (IDP), and Generative AI. Skilled in developing scalable AI solutions across predictive analytics, OCR, NLP, LLMs, Computer Vision, and MLOps, from research and model development to production deployment.

EDUCATION

CADI Ayyad university - National School of Applied Sciences **Safi, Morocco**
Masters degree in and Artificial Intelligence engineering *Sep 2021 – Jun 2026*

Lycée les Champions Safi *Jun 2021*
Baccalaureate in Mathematical Sciences

PROFESSIONAL EXPERIENCE

Mala Inc. **Riyadh, Saudi Arabia**
AI and Risk assessment Engineer — Financial Document Processing

- Design and deployment end-to-end AI-powered Intelligent Document Processing (IDP) solutions for financial documents for BNPL uses, combining OCR, NLP, and Vision-Language Models to automate **KYC and risk assessment** workflows through accurate data extraction, validation, and seamless API integration, significantly improving operational efficiency and data quality.

Attijariwafa Bank Group **Casablanca, Morocco**
Final Year Project (PFE) — AI Engineer Intern

Document Fraud Detection Pipeline for eKYC Identity Verification

- Developed an AI-powered identity verification (eKYC) and fraud detection solution to secure remote bank account opening, simplifying the customer onboarding experience by enabling account creation using only the national ID card—eliminating the need to visit a bank branch.

Intelligent Document Classification

- Developed an AI-powered intelligent document classification and intake system to automate banking back-office and KYC document processing, enabling accurate document routing and seamless integration with core banking systems through REST APIs.

Mass Payment Document Processing

- Designed a scalable AI-powered Document Intelligence pipeline for Mass Payment processing, leveraging layout-aware OCR, NLP, and Vision-Language Models to automate data extraction across 30+ document formats while optimizing production inference through model benchmarking and quantization .

Attijariwafa Bank Group**Casablanca, Morocco***Intelligent Automation Engineer Intern — Cheque Data Extraction*

- Designed a production-ready AI-powered IDP solution for cheque processing, combining OCR engines and multimodal LLMs (OlmOCR, Chandra, GLM) to automate document understanding, achieving a ~70% reduction in manual processing time while supporting optimized on-premise deployment and core banking integration.

OCP Group**Safi, Morocco***AI Engineer Intern — Agricultural Decision Support*

- Developed AI-powered Computer Vision solutions for automated soil and crop analysis, enabling intelligent agricultural decision support through actionable recommendations that optimize farming operations and improve decision-making.

BC-Skills Company**Safi, Morocco***Conversational AI Engineering intern — Virtual Assistant (BC-Skills BOT)*

- Architected and deployed a production Conversational AI chatbot integrating LLMs for multi-turn dialogue; designed full conversation flow and optimised for coherent, context-aware interactions.
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PROJECT EXPERIENCE**AI Portfolio Optimisation Dashboard (QuantCompass)**

- Full-stack AI dashboard for investment portfolio optimisation, financial risk analysis, and performance forecasting with interactive visualisation.

IoT Medical Monitoring Tool

- Real-time monitoring system combining IoT sensors, connected data transmission, and ML-based anomaly detection.

COVID-19 Detection Tool

- Trained and evaluated deep learning models for automated detection from medical images; systematic multi-architecture benchmarking.

AFCON 2025 Morocco Web Platform with Conversational AI

- Full-stack web platform with integrated Conversational AI chatbot; handled frontend (React), backend (Python/FastAPI), and AI layer for team performance analysis and predictions .
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SKILLS**Artificial Intelligence & Machine Learning**

- Machine Learning, Deep Learning, Data Science, Predictive Analytics, Feature Engineering, Model Evaluation & Optimization
- Supervised & Unsupervised Learning, Computer Vision, Natural Language Processing (NLP), Generative AI
- PyTorch, TensorFlow, Scikit-learn, Keras, XGBoost, LightGBM

Document Intelligence & Conversational AI

- Intelligent Document Processing (IDP), OCR, Document Understanding, Information Extraction, Document Classification

- Vision-Language Models (VLMs), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents
- LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, spaCy, PaddleOCR, Tesseract

Model Deployment & MLOps

- Fine-tuning (LoRA, PEFT), Quantization (GGUF, GPTQ, AWQ), On-premise AI Deployment
- REST APIs, FastAPI, Docker, Git, GitHub Actions, MLflow, Weights & Biases

Programming

- Python, SQL, JavaScript, Java, C/C++, Bash, R

Data Engineering & Databases

- Pandas, NumPy, Polars, Apache Spark, Kafka
- PostgreSQL, MySQL, MongoDB

Cloud

- AWS, Google Cloud Platform (GCP)

Languages:

- Arabic (native), French (fluent), English (professional proficiency — written & spoken)
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LEADERSHIP & COMMUNITY

Geeks Club Safi — Organised Python workshops and a Blockchain bootcamp for student engineers

GDSC (Google Developer Student Club) Safi — Completed Salesforce CRM bootcamp

Social Club Safi — One year of structured community leadership